A GROUP SCHEME OVER A FIELD IS SEPARATED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-separated (canonical tag 047G). cited at proof lines 990-992. Reduces separatedness to the identity section being a closed immersion.
- Lemma morphisms-lemma-algebraic-residue-field-extension-closed-point-fibre (canonical tag 01TE). cited at proof lines 992-995. The image of e is a closed point of G.
- Lemma schemes-lemma-characterize-closed-immersions (canonical tag 01LD). cited at proof lines 998-999. Surjectivity of O_G -> e_*O_S gives a closed immersion.

Source reference. Original lemma 047G: sources/stacks-chow/tags.txt, source lines 5507–5507. Cited at proof lines 990-992: reduces the claim to showing e : S -> G is a closed immersion; load-bearing.

Source reference. Original lemma 01TE: sources/stacks-chow/tags.txt, source lines 2371–2371. Cited at proof lines 992-995: the image of e : S -> G is a closed point of G; load-bearing.

Source reference. Original lemma 01LD: sources/stacks-chow/tags.txt, source lines 2082–2082. Cited at proof lines 998-999: O_G -> e_*O_S surjective (skyscraper sheaf) implies e is a closed immersion; load-bearing.

Lemma 0.2. *Let G be a group scheme over a field. Then G is a separated scheme.*

Proof. Say $S = \mathrm{Spec}(k)$ with k a field, and let G be a group scheme over S. By Lemma 047G we have to show that $e : S \to G$ is a closed immersion. By Morphisms, Lemma 01TE the image of $e : S \to G$ is a closed point of G. It is clear that $\mathcal{O}_G \to e_*\mathcal{O}_S$ is surjective, since $e_*\mathcal{O}_S$ is a skyscraper sheaf supported at the neutral element of G with value k. We conclude that e is a closed immersion by Schemes, Lemma 01LD. □

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